

Runxin Yu

Research Interests: Sustainable AI, Climate Policy, Low-Carbon Energy Transition, Resource and Environmental Economics

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Work Experience

Institute of Urban Environment, Chinese Academy of Sciences <i>Assistant Professor</i>	11/2025 - Present
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Education

Ph.D. in Management Science and Engineering, Tsinghua University Supervisor: Prof. Xiliang Zhang	08/2020 - 10/2025
Visiting Scholar, University of California, Berkeley California-China Climate Institute; Supervisors: Fan Dai, Daniel Farber	12/2024 - 06/2025
Bachelor of Management, Sun Yat-sen University	08/2016 - 06/2020

Publications

- [1] **Yu, R.**, Zhu, Z., Qin, S. and Zhang, D., 2026. Dynamic Ex-Post Baselines for Enhancing the Credibility of Improved Forest Management Carbon Offsets. *Environmental Science & Technology*, 60(9), pp.6885-6898.
- [2] Zhu, Z., Mao, H., **Yu, R.** et al., 2026. Integrated planning of net-zero power systems for all. *Nature Energy*, pp.1-21.
- [3] **Yu, R.**, Gordon, J., Dai, F., Lu, S., Zhang, D., He, K., 2026. A comparative review of compliance-linked carbon offset programs in China and California: The new CCER and the Compliance Offset Program. *Advances in Climate Change Research*.
- [4] Zhu, Z., **Yu, R.***, Chen, W.Q. and Zhang, D., 2026. Desert power for the AI era. *Nexus*, 3(2).
- [5] **Yu, R.**, Ma, S., Zhang, D., Zhang, X., 2024. Forest vegetation increased across China's carbon offset projects and positively impacted neighboring areas. *Communications Earth & Environment*, 5, 767.
- [6] **Yu, R.**, Zhang, D., Zhang, X., 2024. Introducing auctioning in China's national carbon market: lessons from international and domestic practices. *Climate Policy*, 1–17.
- [7] **Yu, R.**, Zhang, D., Zhang, X., & Huang, X., 2023. Machine learning for data verification in emissions trading system. *Resources, Conservation and Recycling*, 199, 107239.
- [8] Zhang, X., **Yu, R.***, Karplus, V.J., 2025. The development of China's national carbon market: an overview. *Energy and Climate Management*.
- [9] Zhang, D., **Yu, R.**, Huang, X., & Zhu, K., 2025. A comparative study of energy system transformation toward carbon neutrality in BRICS nations. *Energy and Climate Management*.
- [10] Luo, X., **Yu, R.**, Guo, Y., Jia, H., & Liu, D., 2022. Carbon price prediction of LSTM method based on attention mechanism. *2022 4th International Conference on Smart Power & Internet Energy Systems (SPIES)*, IEEE.
- [11] Zhang, X., Zhang, D., **Yu, R.**, 2021. Theoretical and Practical Insights into China's National Carbon Market Design with Chinese Characteristics. *Management World*, 37(08): 80-95. (Chinese)
- [12] **Yu, R.**, Tan, B., Zhang, D., Lu, S., He, K., 2025. Research on the EU Carbon Market Auction Revenue Management System and Its Implications for China. *Environmental Protection*, 53(21): 35-42. (Chinese)
- [13] **Yu, R.**, Weng, Y., Zhang, X., 2024. International Experience and Reference for Establishing Market Stability Mechanisms in China's National Carbon Market. *Environmental Protection*, 52(13): 68-73. (Chinese)

- [14] Zhang, X., **Yu, R.**, Weng, Y., 2022. Comparison of Carbon Market Institutional Design at Home and Abroad. *Environmental Protection*, 50(22): 16-20. (Chinese)
- [15] Weng, Y., **Yu, R.**, Chen, Y., Zhang, X., 2023. Analysis of Carbon Emission Reduction Cooperation Models Based on Green “Belt and Road” Construction and International Carbon Trading. *Environmental Protection*, 51(24): 72-76. (Chinese)
- [16] Zhang, S., Chen, H., Wang, B., **Yu, R.**, Ma, Z., Miao, Y., 2023. Carbon emission monitoring based on the electricity-carbon relationship of cement enterprises. *China Environmental Science*, 43(07): 3787-3795. (Chinese)

Other Achievements

- Monograph: *Design and Practice of China’s Carbon Market*, main author (in press)
- Monograph: *Analysis of the EU Carbon Market Allowance Management System*, main author (in press)
- Software Copyright: Carbon Market Data Quality Assessment and Early Warning System V1.0
- California-China Carbon Market Joint Research Report: Zhang Xiliang, **Yu Runxin**, Qi Shaozhou, Yang Guangxin: “Theory and Practice of China’s Carbon Emissions Trading System—key issues in China’s national ETS and case study of Hubei pilot ETS”
- EDF Joint Research Report: **Yu Runxin**, Zhang Da, *Research on the Auction Mechanism Design of China’s National Carbon Market* (exhibited at the China Pavilion of COP27)

Research Projects

- [1] **Principal Investigator**, “Design of Ex-Post Dynamic Baselines for Forestry Carbon Sequestration Accounting,” Innovation Fund for Young Scientists, Fujian Natural Science Foundation, 06/2026 – Present.
- [2] **Principal Investigator**, “International Comparison and Policy Implications of Key Mechanisms in China’s National Carbon Market,” Youth Project, Xiamen Natural Science Foundation, 06/2026 – Present.
- [3] **Investigator**, “Research on the Use and Management of Funds from Paid Allocation in China’s National Carbon Market,” Environmental Defense Fund Beijing Office, 12/2024 – Present.
- [4] **Investigator**, “Research on Paid Allocation in China’s National Carbon Emissions Trading Market” (2024WTJF0446), Ministry of Ecology and Environment / Tsinghua Institute for National Governance and Global Governance, 07/2024 – 12/2024.
- [5] **Investigator**, “Research on Fiscal and Tax Policies Supporting ‘Dual Carbon’ Development” (2023WTJF0615), Ministry of Finance / Tsinghua University Think Tank Center, 07/2023 – 10/2023.
- [6] **Investigator**, “Operational Experience of Carbon Accounting in Foreign Financial Institutions and Its Implications for China” (2022WTJF0330), Ministry of Ecology and Environment / Tsinghua Institute for National Governance and Global Governance, 07/2022 – 12/2022.
- [7] **Investigator**, “Application Conditions and Case Studies of Continuous Online Carbon Emission Monitoring in China’s National Carbon Market (Power Sector),” Environmental Defense Fund Beijing Office, 09/2023 – 09/2024.
- [8] **Investigator**, “Analysis of Factors Influencing Carbon Price in China’s National Carbon Market and Research on Auction Mechanism,” Environmental Defense Fund Beijing Office, 09/2022 – 09/2023.
- [9] **Investigator**, “Dialogue on China’s Tradable Performance Standards and China-Japan Carbon Emissions Trading Mechanism,” Toyota Motor Corporation, 04/2022 – 03/2023.
- [10] **Investigator**, “Mechanisms and Empirical Research on Emission Reduction Additionality of Carbon Inclusion,” Beijing Transport Institute, 06/2022 – 06/2023.
- [11] **Investigator**, “Allocation in the Third Compliance Cycle of China’s National Carbon Market and Linkage with the Carbon Emissions Dual Control System,” Environmental Defense Fund Beijing Office, 12/2022 – 12/2023.

- [12] **Investigator**, “Research on the Auction Mechanism Design of China’s National Carbon Market,” Environmental Defense Fund Beijing Office, 09/2021 – 09/2022.
- [13] **Investigator**, “China-California Carbon Market Cooperation Project,” Beijing Society of Innovation and Development Strategy, 06/2021 – 05/2022.
- [14] **Investigator**, “Enterprise Emission Data Quality Analysis and Early Warning System for Carbon Market Participants,” Environmental Defense Fund Beijing Office, 09/2020 – 09/2021.

Professional Practice

Ministry of Finance of the People’s Republic of China 02/2024 - 04/2024

- Member of the special research team of the Department of Natural Resources and Ecological Environment on “Accelerating the Establishment of a Fiscal and Taxation System Adapted to the ‘Dual Carbon’ Goals”

Ministry of Ecology and Environment of the People’s Republic of China 06/2024 - 06/2025

- Assisted the Department of Climate Change in conducting research on key issues in the construction of China’s national carbon market, including allowance auction mechanisms and sector expansion
- Core member of the special working group on central documents related to carbon market construction; participated in drafting the “Opinions of the General Office of the CPC Central Committee and the General Office of the State Council on Promoting Green and Low-Carbon Transition and Strengthening the Construction of the National Carbon Market”

Conferences

- The 7th Workshop on Resource Security and Economic Sciences, Xiamen University, May 23-24, 2026.
- The 25th Annual IEA-IETA-EPRI Workshop on Greenhouse Gas Emissions Trading, Paris, France, December 2-4, 2025.
- The 6th Workshop on Resource Security and Economic Sciences, Xiamen University, May 20-21, 2025.
- The Fifth Iteration of the Workshop in Environmental Economics and Data Science (TWEEDS), Portland, Oregon, April 25-26, 2025.
- The 29th Annual Conference of the European Association of Environmental and Resource Economists (EAERE), Leuven, Belgium, July 1-4, 2024.
- Professional Association for China’s Environment (PACE) Annual Conference 2023: China’s Environment & Development under Carbon Neutrality & De-globalization, Hong Kong, August 18-20, 2023.

Professional Services

- **Referee** for *Climate Policy*, *China Economic Review*, *Energy Economics*, *Climatic Change*, *Nature Humanities and Social Sciences Communications*.
- **Discussant** for EAERE.

University Services

Teaching Assistant

- Graduate Teaching Assistant, “Energy Economics and Policy”, Tsinghua University, Winter 2021
- Undergraduate Teaching Assistant, “Management Accounting” and “Accounting Information Systems”, Sun Yat-sen University, Summer 2019

Student Service

- Class Leader, Institute of Nuclear and New Energy Technology, Tsinghua University, 2022-2023
- Officer, Graduate Students’ Association, organized activities such as the Global University Climate Forum and the Doctoral Students’ Forum, Tsinghua University, 2020-2021

Fellowships and Awards

- Outstanding Student First-class Scholarship, Tsinghua University, 2021-2022 & 2022-2023
- Outstanding Student First-class Scholarship, Sun Yat-sen University, 2019-2020
- Outstanding Student Leader Award, Sun Yat-sen University, 2016-2017

Languages

Chinese (native), English (fluent)

Skills

Microsoft Office Suite, Stata, Python, R